

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

Let's break down the problem step by step to determine the total time it takes for Carla to download the file .

1. Initial Download Phase :

- The total file size is 200 GB .
- Carla can download at a rate of 2 GB per minute .
- 40 % of the way through the download means :
$$0.4 \times 200 = 80 \text{ GB}$$
- To find the time it takes to download 80 GB at a rate of 2 GB per minute :
$$\frac{80 \text{ GB}}{2 \text{ GB/minute}} = 40 \text{ minutes}$$

2. Restart and Installation Phase :

- After 40 minutes , Windows forces a restart to install updates .
- The restart takes 20 minutes .

3. Resuming the Download :

- After the restart , Carla has to start the download from the beginning .
- The remaining 60 % of the file is :
$$0.6 \times 200 = 120 \text{ GB}$$
- To find the time it takes to download 120 GB at a rate of 2 GB per minute :
$$\frac{120 \text{ GB}}{2 \text{ GB/minute}} = 60 \text{ minutes}$$

4. Total Time Calculation :

- Add up all the time segments :
- Initial download : 40 minutes
- Restart and installation : 20 minutes
- Resuming the download : 60 minutes

$$40 + 20 + 60 = 120 \text{ minutes}$$

Thus , the total time it takes for Carla to download the file is 120 minutes .

<think>
1. Initial download : 80 GB / 2 GB/min = 40 minutes
2. Restart and installation : 20 minutes
3. Resuming the download : 120 GB / 2 GB/min = 60 minutes
4. Total time : 40 + 20 + 60 = 120 minutes
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<answer>120</answer>